Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If μ_r and $\in_r$ are the relative permeability and relative permittivity of a medium, its refractive index is given by

(A) $n = \frac{1}{\sqrt{\mu_r \in_r}}$

(B) $n = \sqrt{\mu_r \in_r}$

(C) $n = \frac{1}{\mu_r \in_r}$

(D) $n = \mu_r \in_r$

Q.2

The earth receives energy from the sun at the rate of $1400 \mathrm{Wm}^{-2}$. The total power incident on a roof of dimensions $(16\ \mathrm{m} \times 10\ \mathrm{m})$ is

(A) $8.75\ \mathrm{W}$

(B) $2.80 \times 10^3\ \mathrm{W}$

(C) $0.11\ \mathrm{W}$

(D) $2.24 \times 10^5\ \mathrm{W}$

Q.3

The earth emits radiations in the infrared region of the spectrum. The spectrum is correctly given by

(A) Rayleigh-Jeans law

(B) Planck's law of radiation

(C) Stefan's law of radiation

(D) Wien's law

Q.4

An electromagnetic wave in vacuum has the electric and magnetic field $\vec{E}$ and $\vec{B}$, which are always perpendicular to each other. The direction of polarization is given by $\vec{X}$ and that of wave propagation by $\vec{k}$. Then

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(A) $\vec{X} \| \vec{B}$ and $\vec{k} \| \vec{B} \times \vec{E}$

(B) $\vec{X} \| \vec{E}$ and $\vec{k} \| \vec{E} \times \vec{B}$

(C) $\vec{X} \| \vec{B}$ and $\vec{k} \| \vec{E} \times \vec{B}$

(D) $\vec{X} \| \vec{E}$ and $\vec{k} \| \vec{B} \times \vec{E}$

Q.5

The displacement current flows in the dielectric of a capacitor when the potential difference between its plates

(A) is changing with time

(B) is changing with distance

(C) has assumed a constant value

(D) becomes zero

Q.6

If the frequency of an X-ray is increased by 20%, its energy

(A) increases by 10%

(B) increases by 20%

(C) decreases by 10%

(D) decreases by 20%

Q.7

Which of the following radiations has the least wavelength?

(A) γ-rays

(B) β-rays

(C) α-rays

(D) X-rays

Q.8

When a plane electromagnetic wave travels in a dielectric medium, its electric and magnetic energy densities u_e and u_m are related as

(A) $u_e = cu_m$

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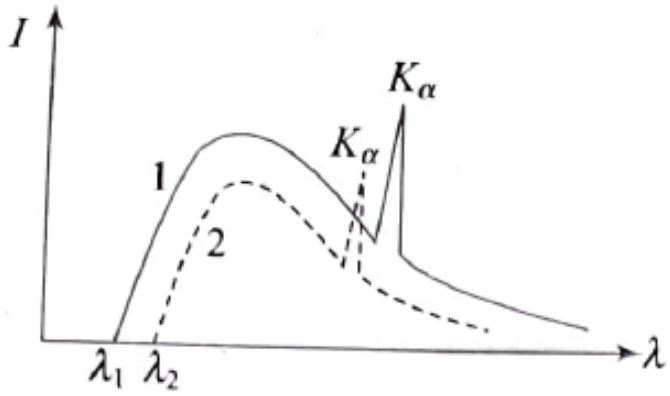
(B) $u_e = \frac{u_m}{c}$

(C) $u_e = u_m$

(D) none of these

Q.9

The intensity distribution of X-rays from two Coolidge tubes operated on different voltages V_1 and V_2 and using different target materials of atomic numbers Z_1 anc Z_2 is shown in the figure. Which one of the following inequalities is true?



(A) $V_1 > V_2, Z_1 < Z_2$

(B) $V_1 > V_2, Z_1 > Z_2$

(C) $V_1 < V_2, Z_1 > Z_2$

(D) $V_1 = V_2, Z_1 < Z_2$

Q.10

Let $\lambda_\alpha \lambda_\beta$ and λ'_α denote the wavelengths of the X-rays of the $K_{\alpha'} K_\beta$ and L_α lines in the characteristic X-rays for a metal

(A) $\lambda_\alpha > \lambda'_\alpha > \lambda_\beta$

(B) $\lambda'_\alpha > \lambda_\beta > \lambda_\alpha$

(C) $\frac{1}{\lambda_\beta} = \frac{1}{\lambda_\alpha} + \frac{1}{\lambda'_\alpha}$

(D) $\frac{1}{\lambda_\alpha} + \frac{1}{\lambda_\beta} = \frac{1}{\lambda'_\alpha}$

Q.11

At a certain instant of time, the potential difference between the plates of a capacitor of capacitance $2\mu\text{F}$ is changing at a rate of $2 \times 10^6\ \text{V s}^{-1}$. The displacement current at that instant is

Q.12

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The K_α X-ray emission line of tungsten occurs at $\lambda = 0.021\ \text{nm}$. What is the energy difference between K and L levels in this atom?

Q.13

The magnetic field in a travelling electromagnetic wave has a peak value of 20nT. the peak value of electric field strength is:

Q.14

The intensity of gamma radiation from a given source is I. On passing through $36\ \text{mm}$ of lead, it is reduced to $I/8$. The thickness of lead, which will reduce the intensity to $I/2$ is

Q.15

A plane electromagnetic were travelling in a nonmagnetic medium is given by
$E = \left(9 \times 10^8 \text{NC}^{-1}\right) \sin\left[\left(9 \times 10^8 \text{rads}^{-1}\right) t - \left(6\ \text{m}^{-1}\right) x\right]$
where x is in metre and t is in second.
The dielectric constant of the medium is

Answers & Solutions

Q.1 Answer:

$n = \sqrt{\mu_r\ \in_r}$

Solution:

$$c = \frac{1}{\sqrt{\mu_0 \in_0}}$$
$$v = \frac{1}{\sqrt{\mu \in}}$$
$$n = \frac{c}{v} = \sqrt{\frac{\mu}{\mu_0}\frac{\epsilon}{\epsilon_0}} = \sqrt{\mu_r\ \in_r}$$

The correct choice is (b).

Q.2 Answer:

$2.24 \times 10^5\ \text{W}$

Solution:

$$P = \left(1400 \text{Wm}^{-2}\right) \times (16 \times 10) \text{m}^2 = 2.24 \times 10^5\ \text{W}$$

which is choice (d).

Q.3 Answer:

Wien's law

Solution:

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The correct choice is (4).

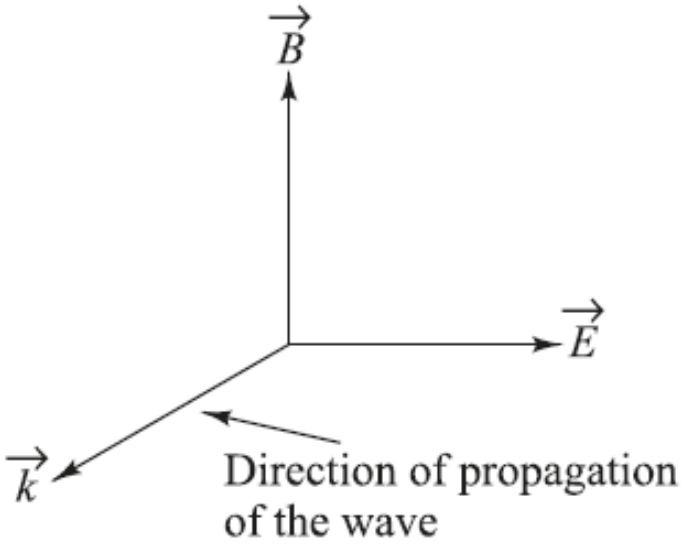
Wien's law states that when the temperature of a black body radiator increases, the overall radiated energy increases and the peak of the radiation curve moves to shorter wavelengths.

Q.4 Answer:

$\vec{X} \| \vec{E}$ and $\vec{k} \| \vec{E} \times \vec{B}$

Solution:

The direction of polarisation vaector $\vec{X}$ is parallel to the plane containing the electric field $\vec{E}$. The propagation vector $\vec{k}$ is perpendicular to $(\vec{E} \times \vec{B})$. So the correct choice is (b).



Q.5 Answer:

is changing with time

Solution:

The displacement current I_d is given by

$I_d = \varepsilon_0 \frac{d\Phi}{d\Phi}$

where Φ is the electric flux. For a parallel plate capacitor

$\Phi = EA$

where A is the area of the plate and E is the electric field between the plates. Now, $E = V/d$ where d is the distance between the plates. Hence

$I_d = \varepsilon_0 \frac{d}{dt}(AE) = \varepsilon_0 A \frac{dE}{dt} = \frac{\varepsilon_0 A}{d} \frac{dV}{dt}$

Thus I_d exists as long as the potential difference V is changing with time t. Hence the correct choice is (a).

Q.6 Answer:

increases by 20%

Solution:

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$E = hv$. **So** $E \propto v$. **Hence if** v **is increased by** 20%, E **will also increase by** 20%. **So the correct choice is (b).**

Q.7 Answer:

γ-rays

Solution:

γ-rays have the highest frequency and hence the shortest wavelength.

Q.8 Answer:

$u_e = u_m$

Solution:

When an electromagnetic wave travels in a dielectric medium, its total energy density is divided equally between its electric and magnetic fields. Hence the correct choice is (c).

Q.9 Answer:

$V_1 > V_2, Z_1 < Z_2$

Solution:

Solution We know that

λa_2

As a for kline,

$\alpha_1 > \lambda_2$

$\Rightarrow x_1 < z_2$

To increase the intensity of

∞ **- rays, voltage is increased.**

Thus,

$\mathrm{I}_2 > \mathrm{I}_2$

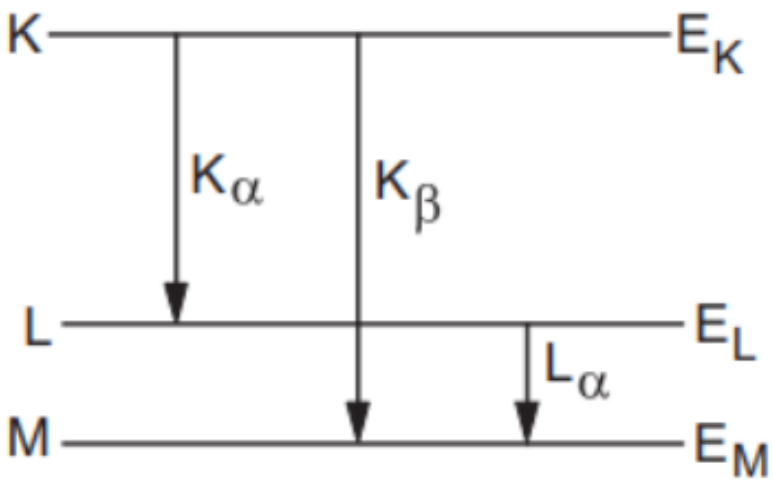
$\rightarrow \mathrm{V}_1 > \mathrm{V}_2$

Q.10 Answer:

$\frac{1}{\lambda_\beta} = \frac{1}{\lambda_\alpha} + \frac{1}{\lambda'_\alpha}$

Solution:

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$E_{\mathrm{K}} - E_{\mathrm{L}} = \frac{hc}{\lambda_\alpha}$
$E_{\mathrm{K}} - E_{\mathrm{M}} = \frac{hc}{\lambda_\beta}$
$E_{\mathrm{L}} - E_{\mathrm{M}} = \frac{hc}{\lambda'_\alpha} = \frac{hc}{\lambda_\beta} - \frac{hc}{\lambda_\alpha}$
or $\frac{1}{\lambda_\beta} = \frac{1}{\lambda_\alpha} + \frac{1}{\lambda'_\alpha}$

Q.11 Answer:

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Solution:

$i_d = \varepsilon_0 \frac{d\Phi_E}{dt}$
$\Phi_E = EA = \frac{VA}{d}$
$\therefore \quad i_d = \varepsilon_0 \frac{d}{dt}\left(\frac{VA}{d}\right) = \frac{\varepsilon_0 A}{d}\frac{dV}{dt}$
$= C\frac{dV}{dt}$
$= 2 \times 10^{-6} \times 2 \times 10^6$
$= 4\ \mathrm{A}$

Q.12 Answer:

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Solution:

The origin of K_α X-ray can be understood on the basis of the Bolr model of atom. If there is only one electron in the K shell, and fill up this vacancy. An electron from higher shell will get de-excited. In this process of de-excitation of electron, a photon is emitted during this transition which is the X-rays.

$$JE = \frac{hc}{\lambda} = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{0.021 \times 10^{-9}}$$
$$= 9.46 \times 10^{-15}\ \mathrm{J}$$
$$= 5.9 \times 10^4 \mathrm{eV} = 59 \mathrm{keV}$$

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Q.13 Answer:

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Solution:

$$E_0 = cB_0$$
$$= \left(3 \times 10^8\right) \times \left(20 \times 10^{-9}\right)$$
$$= 6\mathrm{Vm}^{-1}$$

Q.14 Answer:

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Solution:

The intensity of radiation of intensity I when it passes through a thickness x of an absorbing medium is given by

$I' = Ie^{-\mu x}$ **where μ is a constant. When $x_1 = 36$ m, $I' = \frac{I}{8}$ Therefore,**

$$\frac{I}{8} = Ie^{-\mu x}$$
$$\Rightarrow \quad \frac{1}{8} = e^{-\mu x_1}$$
$$\text{or } e^{\mu x_1} = 8$$
$$\Rightarrow \quad \mu x_1 = \ln(8) = 2.08$$
$$\Rightarrow \quad \mu = \frac{2.08}{x_1}$$

Let x_2 be the thickness of lead for I to reduce to

$I_2 = \frac{I}{2}$

Then

$$\frac{I}{2} = Ie^{-\mu x_2}$$
$$\Rightarrow \quad \mu x_2 = \ln(2) = 0.693$$
$$\Rightarrow \frac{2.08}{x_1} \times x_2 = 0.693$$
$$\Rightarrow \quad x_2 = \frac{0.693 \times x_1}{2.08}$$
$$= \frac{0.693 \times 36 \text{ mm}}{2.08} = 11.99 \text{ mm} \simeq 12 \text{ mm}$$

Q.15 Answer:

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Solution:

$v = \frac{\omega}{k} = \frac{9 \times 10^8}{6} = 1.5 \times 10^8 \text{ ms}^{-1}$

Refractive index $n = \frac{c}{v} = \frac{3 \times 10^8}{1.5 \times 10^8} = 2$

Also $n = \sqrt{\mu_r \in_r}$**. For a non-magnetic medium** $\mu_r = 1$

Therefore

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$n = \sqrt{\epsilon_r} \Rightarrow \epsilon_r = n^2 = (2)^2 = 4$

$\therefore$ **Dielectric constant** $k = \epsilon_r = 4$.

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